6

Acquisition

The present and the following chapters are based on signals recorded according to parameters described in Section 4.3. The theory can be applied similarly to records with a different selection of parameters.

6.1 Motivation

The purpose of acquisition is to determine visible satellites and coarse values of carrier frequency and code phase of the satellite signals.

The satellites are differentiated by the 32 different PRN sequences. The second parameter, code phase, is the time alignment of the PRN code in the current block of data. It is necessary to know the code phase in order to generate a local PRN code that is perfectly aligned with the incoming code. Only when this is the case, the incoming code can be removed from the signal. PRN codes have high correlation only for zero lag. That is, the two signals must be perfectly aligned to remove the incoming code. The third parameter is the carrier frequency, which in case of downconversion corresponds to the IF. The IF should be known from the L1 carrier frequency of 1575.42 MHz and from the mixers in the downconverter. However, the frequency can deviate from the expected value. The line-of-sight velocity of the satellite (with respect to the receiver) causes a *Doppler effect* resulting in a higher or lower frequency. In the worst case, the frequency can deviate up to ± 10 kHz. It is important to know the frequency of the signal to be able to generate a local carrier signal. This signal is used to remove the incoming carrier from the signal. In most cases it is sufficient to search the frequencies such that the maximum error will be less than or equal to 500 Hz; see Akos (1997).

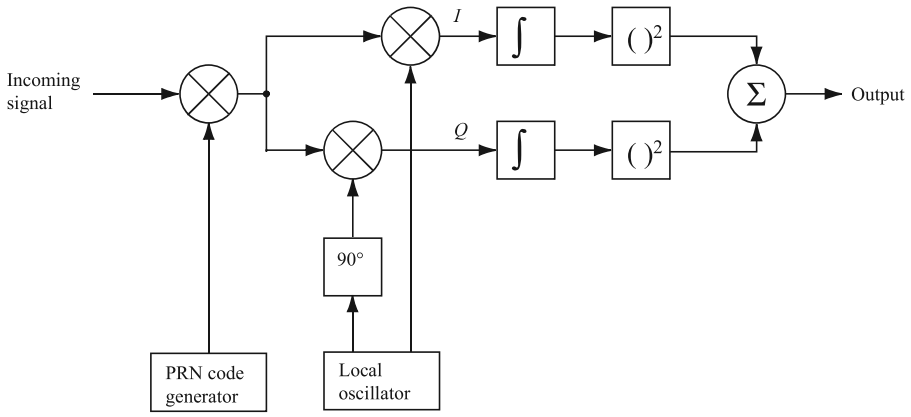


FIGURE 6.1. Block diagram of the serial search algorithm.

In an ordinary receiver, the acquisition is usually performed in an application-specific integrated circuit (ASIC). In a software receiver, it is implemented in software. The following sections describe the theory behind three standard methods of acquisition to demonstrate the possibility of implementing an efficient method in a software receiver.

6.2 Serial Search Acquisition

Serial search acquisition is an often-used method for acquisition in code-division multiple access systems (CDMA). GPS is a CDMA system. Figure 6.1 is a block diagram of the serial search algorithm.

As seen in Figure 6.1, the algorithm is based on multiplication of locally generated PRN code sequences and locally generated carrier signals. The PRN generator generates a PRN sequence corresponding to a specific satellite. The generated sequence has a certain code phase, from 0 to 1022 chips. The incoming signal is initially multiplied by this locally generated PRN sequence. After multiplication with the PRN sequence, the signal is multiplied by a locally generated carrier signal. Multiplication with the locally generated carrier signal generates the in-phase signal I, and multiplication with a 90° phase-shifted version of the locally generated carrier signal generates the quadrature signal Q.

The I and Q signals are integrated over 1 ms, corresponding to the length of one C/A code, and finally squared and added. Ideally, the signal power should be located in the I part of the signal, as the C/A code is only modulated onto that. However, in this case the I signal generated at the satellite does not necessarily correspond to the demodulated I. This is because the phase of the received signal is unknown. So to be certain that the signal is detected, it is necessary to investigate both the I and the Q signal. The output is a value of correlation between the incoming signal and the locally generated signal. If a predefined threshold is exceeded, the frequency and code phase parameters are correct, and the parameters can be passed on to the tracking algorithms.

The serial search algorithm performs two different sweeps: a frequency sweep over all possible carrier frequencies of IF ± 10 kHz in steps of 500 Hz and a code phase sweep over all 1023 different code phases. All in all, this sums up to a total of

$$\underbrace{1023}_{\text{code phases}} \left(\underbrace{2\frac{10{,}000}{500} + 1}_{\text{frequencies}} \right) = 1023 \cdot 41 = 41{,}943 \text{ combinations.} \tag{6.1}$$

Obviously, this is a very large number of combinations. This exhausting search routine also tends to be the main weakness of the serial search acquisition.

The implementation of the serial search acquisition method is very straightforward. The algorithm can be implemented directly based on the block diagram of the method as shown in Figure 6.1.

6.2.1 PRN Sequence Generation

Figure 6.1 shows that the first task in the serial search acquisition method is to multiply the incoming signal with the locally generated PRN sequence. This of course involves the generation of this PRN sequence. Instead of generating PRN sequences every time the algorithm is executed, all possible PRN sequences are generated offline. The 32 different PRN sequences are generated by the PRN generator implemented according to Figure 2.5.

The PRN code generator is implemented using the binary values 0 and 1. However, in the signal processing algorithms it is more convenient to represent the codes with a polar non-return-to-zero representation.

With 32 generated PRN sequences, all possible sequences originating from GPS satellites are created. However, as mentioned in the theory of serial search acquisition, the method involves multiplication with all possible shifted versions of the PRN codes. That is, besides saving the 32 possible PRN codes all possible shifted versions should also be saved. This sums up to a total of 32,736 different PRN codes. To make the multiplication between one of the generated PRN codes and the incoming signal possible, the code has to be *sampled* with 38.192 MHz, like the received signal. This sampling converts the length of a PRN sequence from 1023 to 38,192.

6.2.2 Carrier Generation

The second step is multiplication with a locally generated carrier wave. The carrier generator must generate two carrier signals with a phase difference of 90°, corresponding to a cosine and a sine wave. The carrier must have a frequency corresponding to the IF $\pm$ the frequency step according to the examined frequency area. It must be sampled with the sampling frequency of 38.192 MHz and have a length of 1 ms. A complex signal is generated using the natural exponential function $e^{j2\pi f}$.

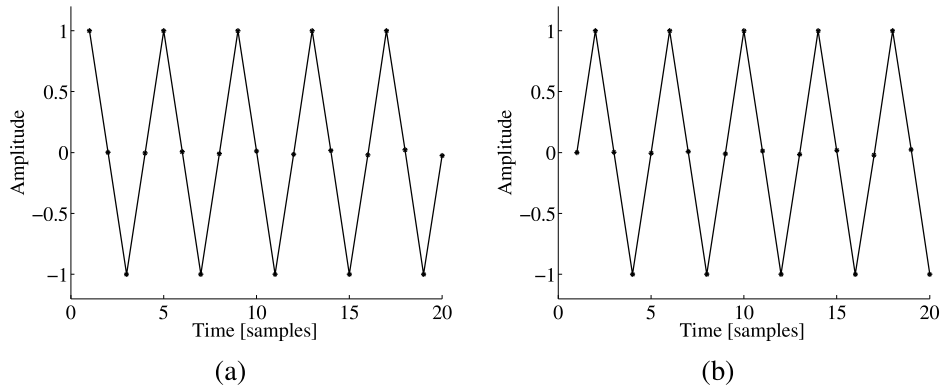


FIGURE 6.2. Locally generated cosine (a) and sine (b) waves.

In MATLAB the 38,192 sample-long sequence, corresponding to 1 ms, is generated as carrier = exp(j * 2 * pi * fc * ts * nn); where fc is the IF plus current frequency offset, ts is the sampling period $t_s = 1/f_s$, and nn is an array with numbers from 0 to 38,191. Now the sine and cosine waves are generated as cosine = real(carrier); and sine = imag(carrier);.

Figure 6.2 shows the first 20 samples of the resulting cosine and sine waves when using a sampling frequency $f_s = 38.192$ MHz and letting $f_c = 9.548$ MHz.

6.2.3 *Integration and Squaring*

The last parts of the serial search algorithm involve an integration and a squaring of the two results of the multiplications with the cosine and sine signals, respectively. The squaring is introduced to obtain the signal power. The integration is simply a summation of all 38,192 points corresponding to the length of the processed data. The squaring is then performed on the result of the summation. The final step is to add the two values from the I arm and the Q arm. If the locally generated code is well aligned with the code in the incoming signal, and the frequency of the locally generated carrier matches the frequency of the incoming signal, the output will be significantly higher than if any of these criteria were not fulfilled. Figure 6.3 shows the output from the serial search acquisition method. Figure 6.3a is the output when performing the acquisition on a satellite that is not visible and Figure 6.3b when performing the acquisition on a satellite that is visible. PRN 19 is arbitrarily chosen as a nonvisible satellite.

6.3 Parallel Frequency Space Search Acquisition

The serial search acquisition method showed that it is a very time-consuming procedure to search sequentially through all possible values of the two parameters frequency and code phase. If any of the two parameters could be eliminated from the search procedure or if possible implemented in parallel, the performance of the procedure would increase significantly.

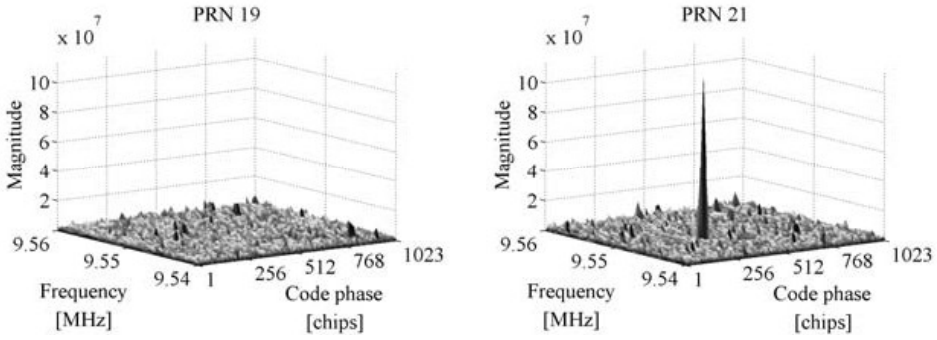


FIGURE 6.3. Output from serial search acquisition. (a) PRN 19 is not visible so no peak is present. (b) PRN 21 is visible so a significant peak is present. The peak occurs at C/A code phase = 359 chips and frequency = 9.5475 MHz.

As the name *parallel frequency space search acquisition* implies, this second method of acquisition parallelizes the search for the one parameter. This method utilizes the Fourier transform to perform a transformation from the time domain into the frequency domain. Figure 6.4 is a block diagram of the parallel frequency space search algorithm.

The incoming signal is multiplied by a locally generated PRN sequence, with a code corresponding to a specific satellite and a code phase between 0 and 1022 chips. The resulting signal is transformed into the frequency domain by a Fourier transform. The Fourier transform could be implemented as a discrete Fourier transform (DFT) or a fast Fourier transform (FFT). The FFT is the faster of the two; but it requires an input sequence with a radix-2 length, that is, 2^n, where n takes positive integer value.

Figure 6.5 illustrates the result of multiplying the incoming signal with a perfectly aligned locally generated PRN sequence. The result is a continuous wave signal. Of course, this only happens when the locally generated PRN code is perfectly aligned with the code in the incoming signal. If the incoming signal contains signal components from other satellites, these components will be minimized as a result of the cross-correlation properties of the PRN sequences.

In parallel frequency space search acquisition, the upper signal in Figure 6.5 is the input to the Fourier transform function. With a perfectly aligned PRN code, the output of the Fourier transform will show a distinct peak in magnitude. The

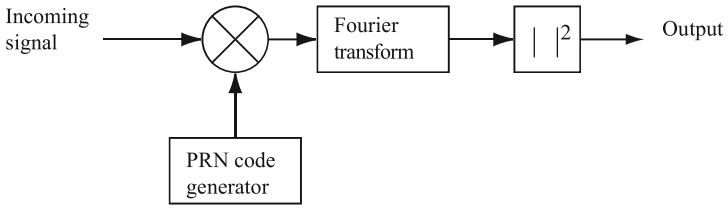


FIGURE 6.4. Block diagram of the parallel frequency space search algorithm.

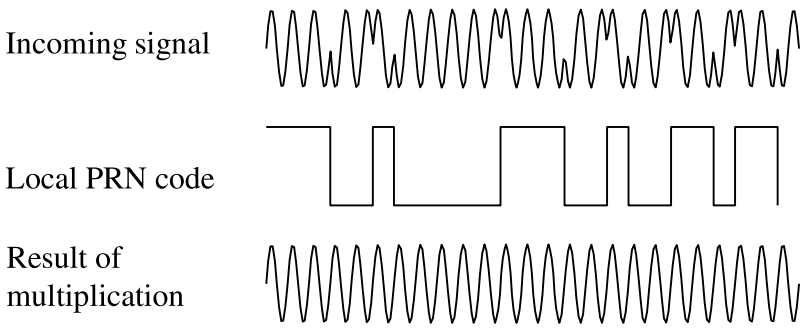


FIGURE 6.5. PRN code demodulation. At the top the PRN code is modulated onto the carrier wave. In the middle is a perfectly aligned PRN sequence. At the bottom we recover the continuous wave after multiplying the incoming signal with the perfectly aligned PRN sequence.

peak will be located at the frequency index corresponding to the frequency of the continuous-wave signal and thereby the frequency of the carrier wave signal.

The accuracy of the determined frequency depends on the length of the DFT. The length corresponds to the number of samples in the analyzed data. If 1 ms of data is analyzed, the number of samples can be found as $1/1000$ of the sampling frequency. That is, if the sampling frequency is $f_s = 10\,\text{MHz}$, the number of samples is $N = 10{,}000$.

With a DFT length of 10,000, the first $N/2$ output samples represent the frequencies from 0 to $\frac{f_s}{2}$ Hz. That is, the frequency resolution of the output is

$$\Delta f = \frac{f_s/2}{N/2} = \frac{f_s}{N}. \tag{6.2}$$

With a sampling frequency of $f_s = 10\,\text{MHz}$ the resulting frequency resolution is

$$\Delta f = \frac{10\,\text{MHz}}{10{,}000} = 1\,\text{kHz}. \tag{6.3}$$

In this case, the accuracy of the estimated carrier frequency is 1 kHz compared to the accuracy of 500 Hz in serial search acquisition.

Figure 6.6 shows the output as two power spectral density (PSD) plots. The PSD algorithm is using FFT for the acquisition. Figure 6.6a is the output of the Fourier transform with a perfectly aligned code phase. This can be noticed from the peak in the plot. Figure 6.6b shows the PSD plot for a not aligned code phase. Note the absence of a peak in the plot.

Where the serial search acquisition method steps through possible code phases and carrier frequencies, the parallel frequency space search acquisition only steps through the 1023 different code phases. This comes with the cost of a frequency domain transformation with each code phase. Depending on the implementation of the frequency domain transformation, it should be possible to make a faster implementation of this method compared to the serial search acquisition method.

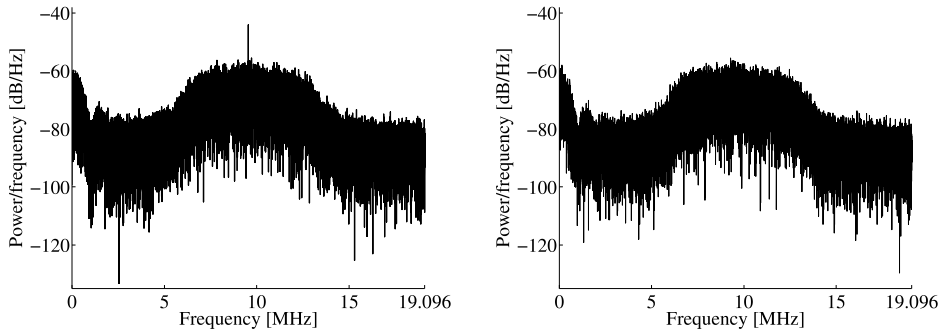


FIGURE 6.6. PSD plot of the incoming signal multiplied by a locally generated PRN code sequence. (a) When multiplying with a perfectly aligned PRN code, the output will show a peak at the carrier frequency. (b) When multiplying with a nonaligned code, the output will not show any peaks. The IF is 9.548 MHz. The Doppler frequency is the same on IF and RF, and thus also the difference between the IF and the peak frequency.

Like the serial search acquisition method, the implementation of the parallel frequency space search method is straightforward. The algorithm can be implemented directly based on the block diagram of the method shown in Figure 6.4.

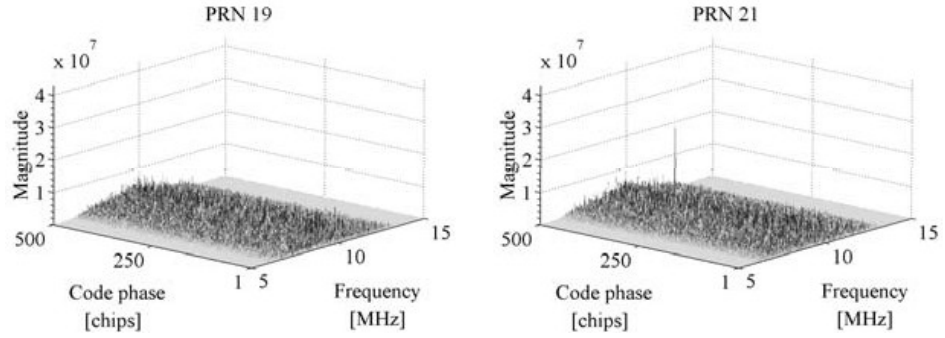
The first part of this method is identical to the first part of the serial search method. That is, a locally generated PRN code must be multiplied with the incoming signal. After the code multiplication, the signal is transformed into the frequency domain through Fourier transform. An efficient tool for that is the fast Fourier transform (FFT). See Oppenheim & Schäfer (1999) for details on the FFT.

After transforming the signal into the frequency domain by means of the FFT algorithm, it becomes a complex signal. If the locally generated code is well aligned with the code in the incoming signal, the output from the FFT will have a peak at the IF plus Doppler offset frequency. To find the possible peak frequency the absolute value of all components are calculated. Figure 6.7 shows the output from the parallel frequency space search method.

6.4 Parallel Code Phase Search Acquisition

As seen from Equation (6.1), the amount of search steps in the code phase dimension is significantly larger than that of the frequency dimension (1023 compared to 41). The previous method parallelized the frequency space search eliminating the necessity of searching through the 41 possible frequencies. If the acquisition could be parallelized in the code phase dimension, only 41 steps should be performed compared to the 1023 in the parallel frequency space search acquisition algorithm.

A recent method in GPS signal acquisition utilizes the before-mentioned advantage of parallelizing the code phase search. This method is simply referred to as parallel code phase search acquisition. In the following we describe the theory behind this method.



(a) (b)

FIGURE 6.7. Output from parallel frequency space search acquisition. The figure only includes the first 500 chip shifts and the frequency band from 5–15 MHz. (a) PRN 19 is not visible so no significant peaks are present in the spectrum. (b) PRN 21 is visible so a significant peak is present in the spectrum. The peak is situated at code phase 359 chips and frequency 9.548 MHz.

The goal of the acquisition is to perform a correlation with the incoming signal and a PRN code. Instead of multiplying the input signal with a PRN code with 1023 different code phases as done in the serial search acquisition method, it is more convenient to make a circular cross correlation between the input and the PRN code without shifted code phase. In the following, a method of performing circular correlation through Fourier transforms will be described; see Oppenheim & Schäfer (1999), page 746, and Tsui (2000), page 140.

The discrete Fourier transforms of the finite length sequences $x(n)$ and $y(n)$ both with length N are computed as

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} \qquad \text{and} \qquad Y(k) = \sum_{n=0}^{N-1} y(n)e^{-j2\pi kn/N}. \quad (6.4)$$

The circular cross-correlation sequence between two finite length sequences $x(n)$ and $y(n)$ both with length N and with periodic repetition is computed as

$$z(n) = \frac{1}{N}\sum_{m=0}^{N-1} x(m)y(m+n) = \frac{1}{N}\sum_{m=0}^{N-1} x(-m)y(m-n). \quad (6.5)$$

In the following we will omit the scaling factor $\frac{1}{N}$.

The discrete N-point Fourier transform of $z(n)$ can be expressed as

$$\begin{aligned} Z(k) &= \sum_{n=0}^{N-1}\sum_{m=0}^{N-1} x(-m)y(m-n)e^{-j2\pi kn/N} \\ &= \sum_{m=0}^{N-1} x(m)e^{j2\pi km/N}\sum_{n=0}^{N-1} y(m+n)e^{-j2\pi k(m+n)/N} = X^*(k)Y(k), \end{aligned} \quad (6.6)$$

where $X^*(k)$ is the complex conjugate of $X(k)$.

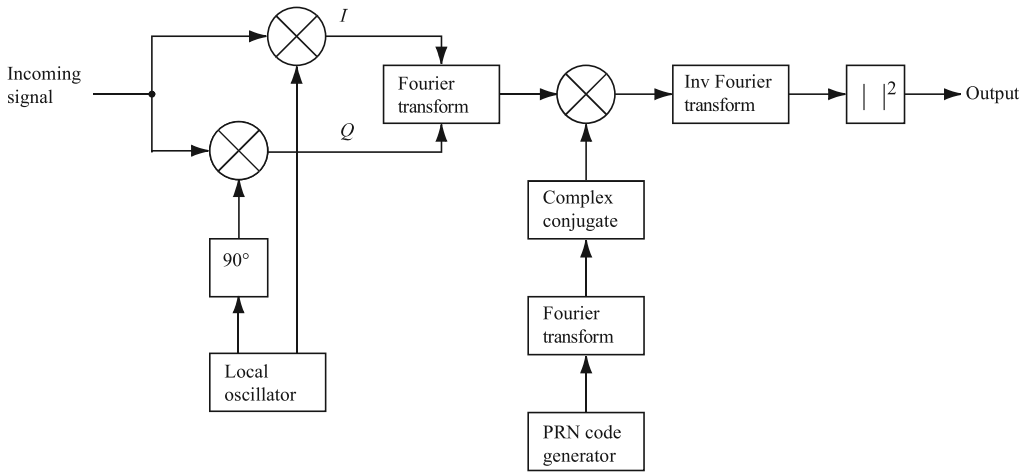


FIGURE 6.8. Block diagram of the parallel code phase search algorithm.

When the frequency domain representation of the cross correlation is found, the time-domain representation can be found through inverse Fourier transform.

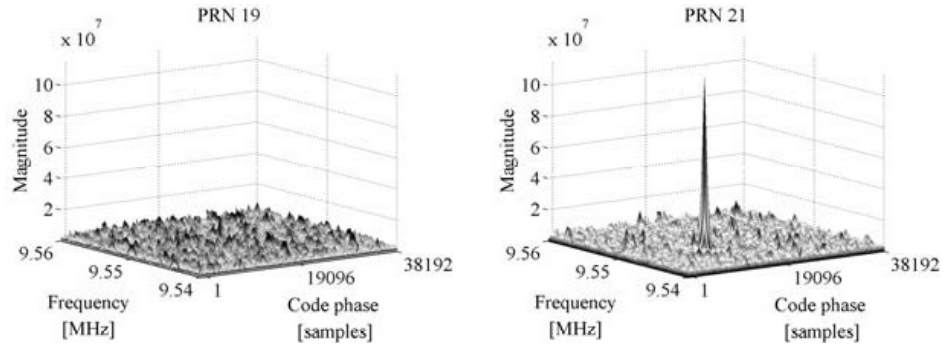
Figure 6.8 is a block diagram of the parallel code phase search algorithm. The incoming signal is multiplied by a locally generated carrier signal. Multiplication with the signal generates the I signal, and multiplication with a 90° phase-shifted version of the signal generates the Q signal. The I and Q signals are combined to form a complex input signal $x(n) = I(n) + jQ(n)$ to the DFT function.

The generated PRN code is transformed into the frequency domain and the result is complex conjugated.

The Fourier transform of the input is multiplied with the Fourier transform of the PRN code. The result of the multiplication is transformed into the time domain by an inverse Fourier transform. The absolute value of the output of the inverse Fourier transform represents the correlation between the input and the PRN code. If a peak is present in the correlation, the index of this peak marks the PRN code phase of the incoming signal.

Compared to the previous acquisition methods, the parallel code phase search acquisition method has cut down the search space to the 41 different carrier frequencies. The Fourier transform of the generated PRN code must only be performed once for each acquisition. For each of the 41 frequencies we perform one Fourier transform and one inverse Fourier transform, so the computational efficiency of the method depends on the implementation of these functions. The accuracy of the parameters estimated by this acquisition method regards the frequency similar to the serial search method. The PRN code phase, however, is more accurate compared to the other methods as it gives a correlation value for each sampled code phase. That is, if the sampling frequency is 10 MHz, a sampled PRN code has 10,000 samples, so the accuracy of the code phase can have 10,000 different values instead of 1023.

In the same way as with the other acquisition methods, the implementation of this one is straightforward, as it can be implemented directly based on the block diagram shown in Figure 6.8.



(a) (b)

FIGURE 6.9. Output from parallel code phase search acquisition. (a) PRN 19 is not visible so no peak is present. (b) PRN 21 is visible so a significant peak is present. The peak is at code phase 13404 samples and frequency 9.5475 MHz.

As seen in Figure 6.8, this method involves almost no new blocks compared to the previous two. As a result of that, many elements can be reused in the implementation. One difference in this method is that only one PRN code should be generated for each acquisition. That is, it is not necessary to take the 1023 different code phases into account.

The first step is to multiply the incoming signal with a locally generated cosine and sine carrier wave, respectively, giving an I and a Q signal component. These two are combined as a complex input to the Fourier transform. The result of the Fourier transform is multiplied with the result of the lower branch of the block diagram in Figure 6.8. This signal is created as follows.

The PRN generator generates a code with no code phase. As in the implementation of the other acquisition algorithms, the code generation is performed *off-line*. The next step performs a Fourier transform of the PRN code, and the result is complex conjugated.

The result of the before-mentioned multiplication is given as input to an inverse Fourier transform implemented as the built-in function IFFT in MATLAB. This function has properties similar to the FFT function regarding execution time.

As mentioned in implementation of the parallel frequency space search acquisition method above, the output from an FFT is complex. This is also the case for the IFFT, so the output needs the same processing as in that case. That is, the absolute value is computed for all components. Figure 6.9 shows the output from the parallel code phase space search method.

6.5 Data Size

The selection of data size used for acquisition can be based on different criteria. The first issue to consider is the effect of navigation data bit transitions. None of the described algorithms ignores these transitions if they occur during the period of acquisition. So to guarantee optimal performance of the acquisition algorithms, it must be ensured that no data transitions occur in the analyzed data sequence.

TABLE 6.1. Execution time for each of the three implemented acquisition algorithms

Algorithm	Execution time	Repetitions	Complexity
Serial search	87	41,943	Low
Parallel frequency space search	10	1023	Medium
Parallel code phase search	1	41	High

As mentioned earlier, the navigation data are transmitted with a rate of 50 bps. This results in possible data bit transitions every 20 ms. If, for instance, 10 ms of data is used for the acquisition algorithm, it might include a bit transition. In fact, there is almost a 50% possibility that it does include a bit transition. (Not exactly 50% because two consecutive data bits might have identical values.) However, if acquisition is performed on two consecutive sequences, each with the length of 10 ms, at least one of the sequences will not include a data transition.

The second issue to consider when selecting the data size used for acquisition is the probability of making a successful acquisition. This issue can be discussed from the idea that the probability of detecting the correct parameters for a certain satellite increases with the amount of analyzed data.

The third issue is the computational demands as a function of the length of data to be analyzed. This is actually the counterpart to the previous issue, as the computations get slower when the sequence gets longer.

The choice of data size used for acquisition has to be a compromise based on the three issues just described. If the issue with data transitions have to be considered it might be necessary to run the acquisition algorithm twice for each acquisition. To ensure good probability of successful acquisition, the data length cannot be too short. However, it cannot be too long because this will cause the computations to be too heavy and time-consuming.

A compromise that will be used involves a data length of 1 ms for the acquisition algorithms. One ms corresponds exactly to the length of one complete C/A code, so it also simplifies the algorithm, making it unnecessary to duplicate the code. The data length can hardly be shorter, because this would involve correlation with an incomplete code. It could be longer, but as mentioned this would decrease the computational performance of the algorithm. To ensure that satellites will be acquired even though a data bit transition occurs in the analyzed data sequence, the algorithm can be run a second time if the first acquisition is unsuccessful.

6.6 Execution Time

As mentioned in the theory behind the acquisition methods, the theoretical performance regarding the computational demands are different between the three methods. So the execution time for each of the three implemented methods will

TABLE 6.2. Results of acquisition of PRN 21 with the three different acquisition algorithms; see captions of Figures 6.3, 6.7, and 6.9.

Search algorithm	**Frequency [MHz]**	**Code phase**
Serial	9.5475	359 chips
Parallel frequency space	9.548	359 chips
Parallel code phase	9.5475	13,404 samples

be measured to be used as a parameter for choosing the right algorithm in the receiver. The execution time is measured using the tic and toc functions in MATLAB. An average PC (Pentium 4, 2.8 GHz) was used for execution time measurements, all measurements are made 10 times, and the mean of these is computed. So the absolute value of the execution times is only approximate as none of these is optimized to run in realtime. The relative measures, however, should indicate which one will have the biggest potential for being implemented in realtime.

The results of the execution-time measurements can be seen in Table 6.1. The table also includes the number of repetitions or combinations the algorithm has to perform and the computational complexity of each of these repetitions.

Note that all relative measures are based on MATLAB performance. Implementation in other languages/environments may change the performance drastically.

6.7 Parameter Estimation

Another parameter that could be used for choosing between the three algorithms is the performance regarding precision of the result. The results from acquisition of satellite 25 using the three acquisition algorithms are shown in Table 6.2.

From Table 6.2 it is evident that all algorithms find the right frequency of the signal. The parallel frequency space search algorithm, however, estimates the frequency to be 9.548 MHz compared to 9.5475 MHz estimated by the other two.

7

Carrier and Code Tracking

7.1 Motivation

The acquisition provides only rough estimates of the frequency and code phase parameters. The main purpose of tracking is to refine these values, keep track, and demodulate the navigation data from the specific satellite (and provide an estimate of the pseudorange). A basic demodulation scheme is shown in Figure 7.1.

The figure shows the scheme used to demodulate the input signal to obtain the navigation message. First, the input signal is multiplied with a carrier replica. This is done to wipe off the carrier wave from the signal. In the next step, the signal is multiplied with a code replica, and the output of this multiplication gives the navigation message. So the tracking module has to generate two replicas, one for the carrier and one for the code, to perfectly track and demodulate the signal of one satellite. In the following, a detailed description of the demodulation scheme is conducted.

7.2 Demodulation

Let f_{L1} and f_{L2} be the carrier frequencies of L1 and L2 for the signal transmitted from satellite k with powers P_C, P_{PL1}, and P_{PL2} for C/A or P code. The C/A code sequence is $C^k(t)$ and the P(Y) code sequence is $P^k(t)$. If the navigation data sequence is called $D^k(t)$, the total signal is given as

$$s^k(t) = \sqrt{2P_C}C^k(t)D^k(t)\cos(2\pi f_{\mathrm{L1}}t) + \sqrt{2P_{\mathrm{PL1}}}P^k(t)D^k(t)\sin(2\pi f_{\mathrm{L1}}t) + \sqrt{2P_{\mathrm{PL2}}}P^k(t)D^k(t)\sin(2\pi f_{\mathrm{L2}}t). \quad (7.1)$$